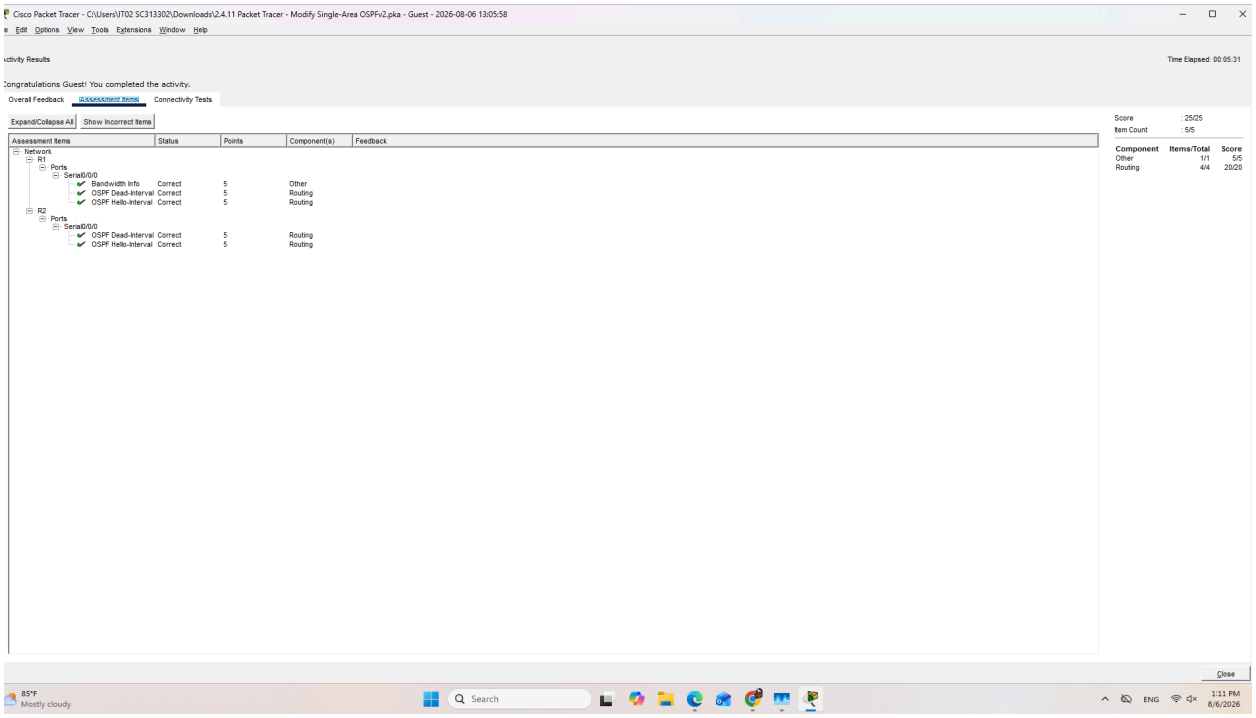
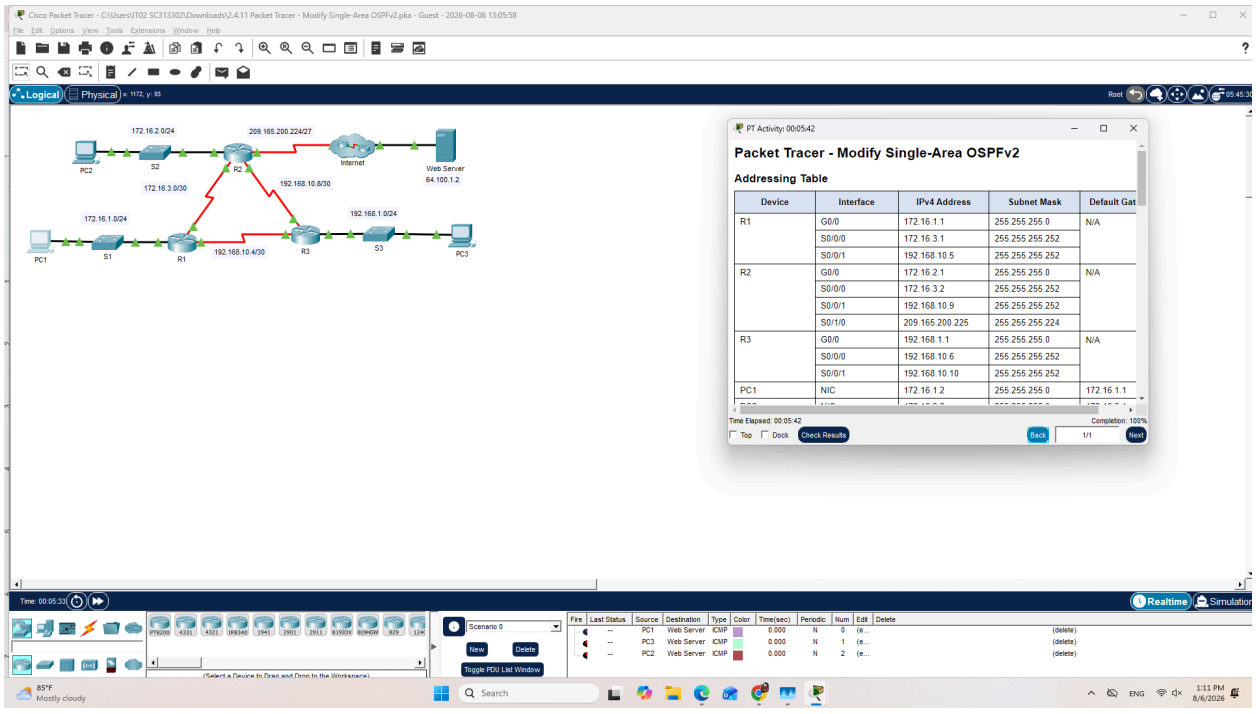




Objective

The objectives of this lab are:

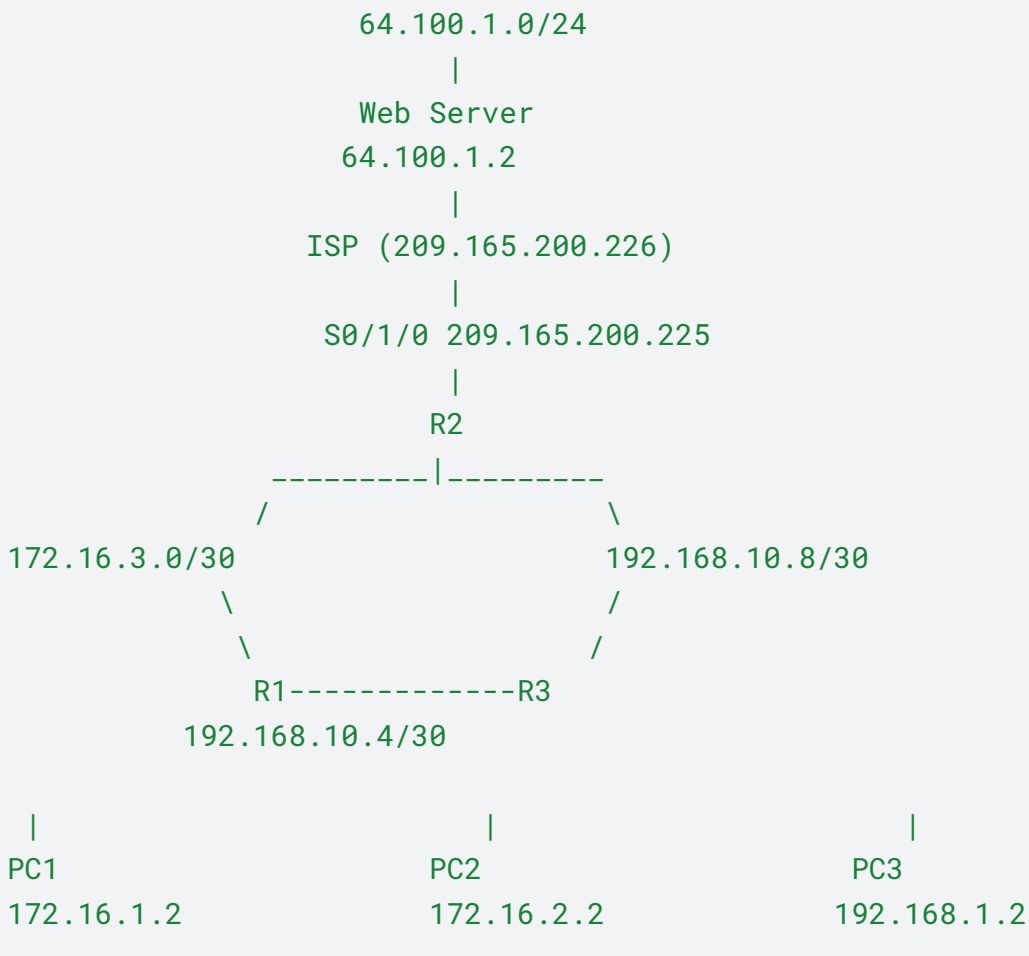
- Modify the default OSPF hello and dead timers.

- Modify interface bandwidth to influence OSPF route selection.
- Verify OSPF neighbor adjacency.
- Verify end-to-end network connectivity after configuration changes.

These objectives are taken directly from the lab instructions.

Network Topology

None



Address Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0	172.16.1.1	255.255.255.0	N/A

R1	S0/0/0	172.16.3.1	255.255.255.252	N/A
R1	S0/0/1	192.168.10.5	255.255.255.252	N/A
R2	G0/0	172.16.2.1	255.255.255.0	N/A
R2	S0/0/0	172.16.3.2	255.255.255.252	N/A
R2	S0/0/1	192.168.10.9	255.255.255.252	N/A
R2	S0/1/0	209.165.200.225	255.255.255.224	N/A
R3	G0/0	192.168.1.1	255.255.255.0	N/A
R3	S0/0/0	192.168.10.6	255.255.255.252	N/A
R3	S0/0/1	192.168.10.10	255.255.255.252	N/A
PC1	NIC	172.16.1.2	255.255.255.0	172.16.1.1
PC2	NIC	172.16.2.2	255.255.255.0	172.16.2.1
PC3	NIC	192.168.1.2	255.255.255.0	192.168.1.1
Web Server	NIC	64.100.1.2	255.255.255.0	64.100.1.1

Router Configurations

Router R1

```
None
enable
configure terminal
```

```
interface Serial0/0/0
 ip ospf hello-interval 15
 ip ospf dead-interval 60
 bandwidth 64

end
write memory
```

Router R2

```
None
enable
configure terminal

interface Serial0/0/0
 ip ospf hello-interval 15
 ip ospf dead-interval 60

end
write memory
```

These are the required configuration changes for the lab.

Verification Commands

Check OSPF Neighbors

```
None
show ip ospf neighbor
```

Expected Result

```
None
Neighbor State : FULL
```

Check OSPF Interface Timers

```
None
show ip ospf interface s0/0/0
```

Expected

```
None
Hello Interval = 15 seconds
Dead Interval = 60 seconds
```

Check Interface Bandwidth

```
None
show interfaces serial0/0/0
```

Expected

```
None
BW 64 Kbit
```

Check Routing Table

```
None
show ip route ospf
```

Expected

OSPF routes are learned successfully.

Connectivity Test

Ping Results

Source	Destination	Result
PC1	PC2	Success
PC1	PC3	Success
PC1	Web Server	Success
PC2	PC3	Success
PC2	Web Server	Success
PC3	Web Server	Success

The lab requires verifying that all PCs can ping one another and the web server after the configuration changes.

Traceroute Results

Before Changing Bandwidth

```
None
PC1
 ↓
R1
 ↓
R2
 ↓
ISP
```

```
 ↓
Web Server
```

Example:

```
None
172.16.1.1
172.16.3.2
209.165.200.226
64.100.1.2
```

After Setting Bandwidth = 64 kbps

```
None
PC1
 ↓
R1
 ↓
R3
 ↓
R2
 ↓
ISP
 ↓
Web Server
```

Example:

```
None
172.16.1.1
192.168.10.6
172.16.3.2
209.165.200.226
64.100.1.2
```

Questions and Answers

1. Why did the OSPF neighbor relationship fail after changing the hello and dead timers on R1?

Answer:
Because the hello and dead timer values no longer matched those configured on R2. OSPF neighbors must use identical hello and dead intervals to maintain adjacency.

2. How was the OSPF adjacency restored?

Answer:
By configuring the same hello interval (15 seconds) and dead interval (60 seconds) on R2's Serial0/0/0 interface, allowing the routers to re-establish a FULL neighbor relationship.

3. Why did the traceroute path change after modifying the bandwidth?

Answer:
Changing the interface bandwidth altered the OSPF cost of the R1–R2 serial link. Since OSPF selects the path with the lowest total cost, traffic was redirected through R3 instead of the direct R1–R2 link.

4. Does changing the bandwidth command change the actual speed of the serial interface?

Answer:
No. The `bandwidth` command only changes the bandwidth value used by routing protocols (such as OSPF) to calculate routing metrics. It does not change the physical transmission speed of the interface.

Conclusion

In this lab, the OSPF hello and dead timers were successfully modified on both R1 and R2 to restore neighbor adjacency. The bandwidth on the R1 Serial0/0/0 interface was then reduced to 64 kbps, causing OSPF to recalculate path costs and choose an alternate route through R3. After these modifications, all PCs maintained full connectivity with one another and with the web server, demonstrating correct OSPF operation and dynamic route selection.